

Name: caleb, vincent, collin, zoe _____ Date: 2/20/26 _____ Class: _____

MyDesign Portfolio

Element E Initial Template - AT - 15 points

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

This section should describe how your proposed solution is supported by STEM (science, technology, engineering, mathematics, computer science, etc.) concepts.

1. Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.
2. Be sure to apply the applicable STEM concepts in context, directly to the specifics of your design. Review concepts learned in your other classes and tackle new concepts as appropriate. Include any relevant calculations.
3. Element C indicates that you should have at least three high-priority design requirements to receive a score of 3 (and four for a score of 4, and five for a 5). Your number of design requirements and your number of high-priority design requirements should be the same throughout your portfolio. You should use the same three to five design requirements here too, and apply **at least three of them** to STEM concepts to achieve a score of 5 on E1. Use as many copies of this sub-template as needed.

High-Priority Design Requirement #1 from Element C - To ensure the safety of all spectators, athletes, and coaches, the lights at Tyoka Jackson field must be bright enough for practices and games.

STEM Concepts Involved in the requirement: brightness quality, safety, and the durability of the lights.

Describe how STEM concepts apply directly to the specifics of your design requirement: Stem concepts apply to this design because the lights must produce a lot of energy and lighting up the field which requires math for the distance.

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High-Priority Design Requirement #2 from Element C - the lights must be safe and energy efficient to meet school and county standards. Safety rules and energy efficiency have always been a concern for the school community and neighboring residents, especially regarding glare and power use.

STEM Concepts Involved in the requirement: we will be using mathematical equations and modeling

Describe how STEM concepts apply directly to the specifics of your design requirement: Electrical power will require math and to calculate the energy needed for each light so it can supply enough energy. So the equation would be like $\text{power} \times \text{time} = \text{energy}$

High-Priority Design Requirement #3 from Element C - Adjustable lights help control energy consumption and help reduce how much residents see light at night.

STEM Concepts Involved in the requirement: Ohms law, and electrical safety so if anything happens we will be able to fix it. Also having balance.

Describe how STEM concepts apply directly to the specifics of your design requirement: We do not want the lights to fall while we adjust them so we must calculate how much force will be used.

High-Priority Design Requirement #4 from Element C (if you have one)- The lights must fit in the stadium without interfering with bleachers, walkways, or nearby homes. Space and zoning issues have caused disagreements in the past with neighbors, so placement matters

STEM Concepts Involved in the requirement: weatherproofing, materials

Describe how STEM concepts apply directly to the specifics of your design requirement: We will need to find the durability for wind and being water proof so the lights don't cut out and so that no electrical components get damaged.



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Proposed Solution to STEM Principles and Practices (E2)

While E1 was focused on how STEM applies to the design requirements, E2 is focused on how STEM is used to support the design solution.

For E2, use STEM concepts to justify your design decisions. It's important that the STEM Principles and Practices that are discussed are actually appropriate and useful to the problem. Use as many copies of this sub-template as needed.

STEM Principles for Design Solution #1

STEM Concept: . Energy being stored and the power systems so electrical engineering.

Describe how the STEM concept applies directly to the specifics of the first solution to your problem: $\text{power} \times \text{time} = \text{energy}$ this equation can help because the lights must last during the practice, or game and doing this math can help with safety not over doing anything or under doing it.



You can only work on the next sections after you have built and tested your first prototype. You need to skip to the "Evidence is Reviewed" section.

STEM Principles for Design Solution #2

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your solution:



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We must research and calculate how much light we need to light up the field at different times of the day. This requires us to calculate how much lumens and watts we need for this. We need to use Om's law, Cosine Law, and Inverse square law.

STEM Principles for Design Solution #3 (if completed)

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your solution:
We must figure out what angle the light fixtures must be at to produce optimal performance and satisfaction for stakeholders. We must use geometry and tools such as protractors to determine this.



Evidence Design is Reviewed (E3)

"Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

E3 is about experts reviewing what you've said about the STEM connections in E1 and E2. It's important here to talk to an appropriate field expert who knows about the STEM content of the project - not to a stakeholder.

Documentation #1 (Only one documentation section needed for full credit):

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What format: Interview notes, emails, audio/video tape with transcript, etc.

- What the expert said about the connection you've made from your design requirements to STEM principles and practices: *Approval: try looking into how much light is considered for the neighborhood and look into the legal side of it*
- *To try to get best lighting is to go look at other stadiums to see how their lights are set up, how many they have, and where they are positioned*
- What the expert said about how STEM principles and practices substantiate your design solution: *Justin talked about how getting the measurements of the field and track can be a major part for the project because it will give us a good estimate for where we can put our lights on the football field and track.*

Documentation #2 (+ making only):

What format: Interview notes, emails, audio/video tape with transcript, etc.

- *Think about the cost/ budget and try to propose something to admin and have a budget set (good for them to hear)*
- What the expert said about the connection you've made from your design requirements to STEM principles and practices
- What the expert said about how STEM principles and practices substantiate your design solution:

Documentation #3 (+ making only):

What format: Interview notes, emails, audio/video tape with transcript, etc. *Sell idea and think about who we are selling it to and think about the concerns they might have, and prove why it's a good idea*

- What the expert said about the connection you've made from your design requirements to STEM principles and practices:



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- What the expert said about how STEM principles and practices substantiate your design solution:

Experts with Credentials (E4)

Document who your field experts are (names are not necessary) and what qualifies them as an experts. Use as many copies of this sub-template as needed.

Expert #1 (from documentation 1): Describe credential(s) and what makes them qualified as an expert. - Justin Bechet is a Controls Design Engineer at Micron Technology and went to Kennesaw State University

Expert #2 (from documentation 2): Describe credential(s) and what makes them qualified as an expert. - He has his bachelors degree of Engineering, Mechatronics, Robotics, and Automation Engineering.

Expert #3 (from documentation 3): Describe credential(s) and what makes them qualified as an expert. - He is a recent graduate and now has a job as a Controls Design Engineer at Micron Technology.

Verification or Corrective Response to Design Review (E5)

Your conversation/interview with the expert(s) should inform some corrective responses. In E5, describe how those reviews informed your responses.

For example,

"_____ told us _to_, which informed our decision to change/reconsider ____." (or "...which verified/reinforced our decision to ____.")

Make sure to document the experts' reviews (suggestions/feedback or criticisms of your proposed solution) and how they will inform iterations you will make to your proposed solution. Use as many copies of this sub-template as needed.

Interview with Expert #1 informs corrective response: "_justin____ told us _to_ check the county requirements_, which informed our decision to change/reconsider __the design of the light fixtures__." (or "...which verified/reinforced our decision to



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Interview with Expert #2 informs corrective response: Justin Bechet told us to try to get best lighting is to go look at other stadiums to see how their lights are set up. Which verified our decision to look at deigns and think more about the size of the lights and how wide they should be.

Interview with Expert #3 informs corrective response:

	Criteria	5	4	3	2	1	0
	3. Evidence that Design is Reviewed. There is _____ evidence that the application of STEM principles & practices (or suitable alternative) has been reviewed by expert(s).	<i>substantial</i>	<i>adequate</i>	<i>some</i>	<i>minimal</i>	<i>no - but review was requested -</i>	<i>no</i>
	4.Design Reviewed by Experts. _____ expert(s) have/has reviewed the design for application of STEM principles & practices.	<i>Two or more, with at least two with documente d, relevant credentials</i>	<i>Two or more, with at least one with documente d, relevant credentials</i>	<i>At least one with documented, clearly relevant credentials</i>	<i>At least one with documented credentials</i>	<i>Only one with unclear credentials</i>	<i>No</i>

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5.Design Review Results in Verification. The design reviews provided _____.	<i>clear and detailed confirmation (verification) or substantial detail necessary to inform a corrective response</i>	<i>clear confirmation (verification) or detail necessary to inform a corrective response</i>	<i>partial confirmation (verification) or some detail necessary to inform a corrective response</i>	<i>does not provide clear confirmation (verification) or suggestions for a corrective response, but at least some response is provided</i>	show no evidence of confirmation (verification) or suggestions for a corrective response by an expert, but includes notes of attempts to gain feedback	<i>show no evidence of confirmation (verification) or suggestions for a corrective response by an expert</i>
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